

AI Guidebook - Test Plan

1. Test Case Table

Test ID	Test Type	Input / Trigger	Expected Output	Test Method	Pass / Fail Criteria	Notes
TC-01	Unit	Create a new UsageLog object with valid tool name, description, and assignment ID.	UsageLog object is validated and saved successfully.	Jest with mongodb-memory-server	Pass: Document is saved and can be retrieved. Fail: Save operation fails or throws an error.	Tests core UsageLog schema (FR2).
TC-02	Unit	Create a new UsageLog object missing a required field like toolName.	Validation error is thrown.	Jest with mongodb-memory-server	Pass: Validation error is thrown. Fail: Object is saved despite missing field.	Ensures schema integrity and data quality.
TC-03	Unit	Execute complianceService.checkLog(logData, courseRules) with a valid log that violates a rule like "full generation".	A violation object is returned, correctly identifying the triggered rule.	Jest unit test for complianceService.js	Pass: Violation object matches expected rule ID and message. Fail: No violation or incorrect violation returned.	Tests core compliance logic (FR5).
TC-04	Unit	Execute complianceService.checkLog with a malformed rule JSON in the database.	Service handles the error gracefully, returning a generic "rule unavailable" message without crashing.	Jest unit test with mock DB returning invalid JSON	Pass: Service returns a fallback response and does not throw an unhandled exception. Fail: Service throws an exception or crashes.	Tests system resilience to misconfiguration (NFR5).

TC-05	Integration	Send a POST /api/logs request with a valid JSON body containing log data.	1. HTTP response status 201 Created. 2. The log is successfully stored in the database.	Supertest (API endpoint test)	Pass: Status is 201 and database contains the new log. Fail: Incorrect status or log not found in DB.	Validates full API endpoint for logging (FR2).
TC-06	Integration	Make a request that triggers the autoLoggingMiddleware. Simulate a delay in the logging service.	The HTTP response is sent back to the client <i>before</i> the log is fully written to the database.	Supertest with spies/mocks on loggingService.js	Pass: Response is received before the async logging operation completes. Fail: Response is blocked until logging finishes.	Verifies non-blocking behavior (FR4, NFR4).
TC-07	Integration	Send a POST /api/logs request that triggers a compliance rule violation (based on course rules).	The API response includes a compliance object containing the expected violation details (FR6).	Supertest with seeded policy rules in DB	Pass: Response body contains compliance violations with correct rule ID and educational message. Fail: Compliance data missing or incorrect.	End-to-end API test from log creation to compliance check (FR5, FR6).
TC-08	Integration	Send a DELETE /api/logs/user/{userId} request.	1. HTTP response status 200 OK. 2. All logs for the specified user are removed from the DB. 3. Logs for other users remain untouched.	Supertest + DB verification	Pass: Only the target user's logs are deleted. Fail: Logs of other users are affected or deletion fails.	Verifies GDPR-compliant data deletion (NFR1).

TC-09	System (E2E)	User journey: Student navigates to the form, fills it out, and submits their AI usage.	1. Form submits successfully. 2. The compliance panel displays feedback based on the log.	Cypress E2E test	Pass: UI shows success state and correct compliance feedback. Fail: Form error, or compliance feedback is incorrect/not displayed.	Tests the core student workflow (FR2, FR3, FR6).
TC-10	System (E2E)	User journey: Student logs a permissible AI interaction like, “edited for grammar”.	The ComplianceFeedbackPanel displays an educational message but no violations.	Cypress UI test	Pass: Panel shows a positive/neutral educational message and no violation list. Fail: A violation is incorrectly shown.	Tests educational feedback priority for compliant usage (FR6, NFR6).
TC-11	System (E2E)	User journey: Student submits a log that violates a major rule like “full generation”.	The ComplianceFeedbackPanel displays the educational message first, followed by a clear violation with a suggested action.	Cypress UI test	Pass: Violation details and suggested actions are rendered correctly. Fail: Violation is not shown or is unclear.	Tests clarity and actionability of warnings (FR5, FR6, NFR3).
TC-12	System (Integration)	Render the main application page for a specific assignment.	The EthicalGuidanceCard component is visible and displays proactive, context-relevant guidance.	React Testing Library / Component test	Pass: Component is in the DOM and contains non-empty guidance text. Fail: Component is missing or empty.	Verifies proactive display of ethical guidance (FR7).

TC-13	NFR (GDPR)	Make an authenticated DELETE /api/logs/user/{userId} request. Then, attempt to query the database directly for those logs.	1. Request succeeds. 2. Subsequent direct DB queries return no documents for that user.	Automated API test + manual DB verification	Pass: Logs are permanently removed. Fail: Logs are still present or only marked as deleted (if soft-delete not intended).	Confirms permanent data erasure per GDPR “right to be forgotten” (NFR1).
TC-14	NFR (Performance)	Simulate 100 concurrent virtual users making requests that trigger the auto-logging middleware.	Average response time for the requests does not increase significantly compared to a baseline without logging (threshold: < 5% increase).	Load test (k6 / Artillery / JMeter)	Pass: Response time increase is within the defined threshold. Fail: Response time degrades significantly.	Validates non-blocking, high-performance logging (NFR4).
TC-15	NFR (Usability)	Present a student user with a compliance warning generated by the system (like., from TC-11).	User can correctly identify the rule they broke, understand why it's a rule, and state the suggested next action.	Usability testing / Heuristic evaluation like with 3-5 users	Pass: >80% of users can correctly interpret the message and action. Fail: Users are confused or misinterpret the feedback.	Ensures warnings are clear, educational, and actionable (NFR3, NFR6).

TC-16	NFR (Configurability)	1. Update a course's PolicyRule JSON in the database. 2. Trigger a compliance check for a log in that course.	The compliance check uses the <i>updated</i> rule, not the old one. No code redeployment is performed.	Database update + API test	Pass: Compliance result reflects the new rule. Fail: The system continues to use the old rule or requires a restart.	Verifies runtime configurability of rules (NFR5).
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2. Coverage Definition

- **Code Coverage Goals:**

- **Line of Code (LOC) Coverage:** $\geq 60\%$ for the backend services (services/, middleware/) and $\geq 50\%$ for frontend components.
- **Branch/Path Coverage:** $\geq 70\%$ for all conditional logic within the complianceService.js (rule evaluation, error handling) and autoLoggingMiddleware.js (error isolation paths).

- **Requirement Coverage Goal:**

- **Functional Requirement Coverage:** 80% of FR2-FR7 must be directly linked to one or more passing test cases.

- **Non-Functional Requirement (NFR) Coverage Goal:**

- **100% of NFR Categories:** Each identified NFR (GDPR, Performance, Usability, Configurability, Educational Priority) must be validated by at least one dedicated test, as shown in the traceability matrix.

3. Traceability Matrix

Requirement ID	Requirement Description	Test Case(s)	Implementation Reference
FR2	Student documentation of AI tools	TC-01, TC-05, TC-09	models/UsageLog.js, routes/logRoutes.js, components/UsageLogForm.jsx
FR3	Concise structured documentation	TC-09	components/UsageLogForm.jsx (UI/UX design)
FR4	Automatic logging of AI interactions	TC-06, TC-14	middleware/autoLoggingMiddleware.js, services/loggingService.js
FR5	Policy compliance checking	TC-03, TC-07, TC-11	services/complianceService.js, models/PolicyRule.js
FR6	Clear compliance explanations	TC-07, TC-10, TC-11, TC-15	components/ComplianceFeedbackPanel.jsx, services/complianceService.js
FR7	Ethical guideline feedback	TC-12	components/EthicalGuidanceCard.jsx

NFR1	GDPR compliance (data ownership & deletion)	TC-08, TC-13	routes/logRoutes.js (DELETE endpoint), models/UsageLog.js (user ID field)
NFR3	Detailed, actionable compliance warnings	TC-11, TC-15	components/ComplianceFeedbackPanel.jsx
NFR4	Non-blocking background operation	TC-06, TC-14	middleware/autoLoggingMiddleware.js (async pattern)
NFR5	Course-specific configurable rules	TC-04, TC-16	models/PolicyRule.js, services/complianceService.js (rule loading logic)
NFR6	Educational priority in feedback	TC-10, TC-15	components/ComplianceFeedbackPanel.jsx (rendering logic)